

ECHS 5-PATTERN FORENSIC LEAKAGE SYSTEM

Hinglish Explanations of Fraud Patterns & Auditing Recommendations

This explanations guide details the **5-Pattern Behavioral Taxonomy** implemented for ECHS Module 20 forensic leakage detection. It provides clear, executive-level definitions of each fraud mechanism (**Kya Hai**) vs. normal clinical behavior (**Kya Nahi Hai**) along with verified case counts and financial statistics.

1. Key Words Aur Definitions (Hinglish)

- **Prevented Leakage (Audit Deductions):** Yeh woh paisa hai jo ECHS ne bills verification ke dauran pakad kar kaat liya (total **₹3,612.41 Cr**). Yeh actual savings hai, yaani yeh paise hospitals ko nahi diye gaye.
- **Realized Leakage (Approved Fraud):** Yeh woh amount hai jo suspicious patterns hone ke bawajood system me auto-approve ho gaya aur hospitals ko pay kar diya gaya (total **₹3,225.01 Cr**). Yeh ECHS budget ka direct financial loss hai.

2. 5 Fraud Patterns Ki Detailed Explanation

Pattern 1: Beneficiary & Dependency Abuse

Smart cards ke profile data me falsification ya details lend karke non-entitled members ko bill karna.

1a. Card Sharing (Concurrent Admissions)

Kya Hai: Ek hi smart card ko multiple families share karti hain aur same day do alag-alag hospitals me admissions/treatments ke liye use karti hain.

Kya Nahi Hai: Yeh genuine medical transfers ya emergency shifting nahi hai jahan patient ek hospital se dusre hospital proper documentation se shifted hota hai.

Data Metrics: 11,137 pairs of concurrent admissions | **₹118.78 Cr** Realized Leakage

1b. Demographic Profile Mismatch

Kya Hai: Identity chupani ya non-entitled logon ka treatment bill karne ke liye gender-relationship conflict code daalna (e.g. Male cardholder listed as 'Wife').

Kya Nahi Hai: Yeh database entry ki simple spelling ya single clerical key-press error nahi hai.

Data Metrics: 4,221 claims mismatch | **₹24.40 Cr** Realized Leakage

1c. Deceased Beneficiary Lazarus Billing

Kya Hai: Cardholder ki death ho chuki hai par card active hai, aur death date ke baad ki hospital admissions bill ho rahi hain.

Kya Nahi Hai: Yeh death bed treatment ya critical emergency bills nahi hain jo patient ki death se pehle perform hue.

Data Metrics: 524 claims | **₹0.64 Cr** Realized Leakage

Pattern 2: Systematic Provider Abuse & Collusion

Hospitals dwara systematic overbilling karna ya fake referral networks run karna.

2a. Hospital Audit Leakage Ranking

Kya Hai: Kuch empanelled hospitals systematically packages rates ko expand karte hain, jisse inka bill audit me har baar 25%+ reject hota hai.

Kya Nahi Hai: Yeh pre-agreed rates ya emergency categories ke generic standard deductions nahi hain.

Data Metrics: 4,000 hospitals ranked | **₹32,649.04 Cr** total approved ECHS budget scanned

2b. Referral Collusion Rings (Doctor-Hospital Loop)

Kya Hai: Ek single doctor aur hospital ka referral ring, jo same patients ko multiple times high-value claims loop me daalta hai (Index > 2.0).

Kya Nahi Hai: Yeh dialysis, chemo, ya genuine chronic care treatments ke cycles nahi hain.

Data Metrics: 1,111 doctor-hospital rings detected (338,280 claims loop) | **₹2,700.74 Cr** approved

Pattern 3: Doctor-Level & Clinical Anomaly

Surgeons ke clinic credentials ka misuse karke phantom ya cloned billing karna.

3a. Superman Surgeons (Surgeon Cloning)

Kya Hai: Senior specialist doctor ke credentials misuse karna aur unke name par ek hi din me 15+ complex surgeries bill karna (clinically impossible).

Kya Nahi Hai: OPD clinic visits me 15 se 20 normal checks karna surgeon cloning nahi hai; yeh theatre (OT) operation fraud hai.

Data Metrics: 2,193 doctor-days flagged (47,433 claims) | **₹265.97 Cr** approved

Pattern 4: Claim Manipulation & Billing Evasion

Audits aur financial powers validation limits se bachne ke liye tricks use karna.

4a. CFA Threshold Avoidance (₹99k Trick)

Kya Hai: ₹1 Lakh ke manual review vetting block se bachne ke liye bill values ko intentionally ₹99,000 se ₹99,999 me split/limit karna.

Kya Nahi Hai: Yeh package price change ya random clinical billing cost fluctuation nahi hai.

Data Metrics: 8,146 claims exactly in this range | **₹68.13 Cr** Realized Leakage

4b. Duplicate Claims (Split Billing)

Kya Hai: Same patient, same hospital aur same amount ka bill 48 hours ke andar do baar upload karke double payment nikalna.

Kya Nahi Hai: Multi-stage clinical step invoicing nahi hai jahan details separate code se aati hain.

Data Metrics: 336,512 duplicate pairs | **₹351.02 Cr** Realized Leakage

Pattern 5: Temporal Surge & Referral Bypass

Bypass channels aur month-end target billing periods ka time abuse karna.

5a. Weekend Emergency Surge

Kya Hai: Polyclinic referral letters se bachne ke liye planned elective admissions ko Friday/Saturday/Sunday emergency routes par dikhana.

Kya Nahi Hai: Trauma, sudden heart attacks, ya real emergency cases ke weekend admissions nahi hain.

Data Metrics: 830,866 claims weekend admissions | █4,751.76 Cr Realized Leakage

5b. Month-End Billing Spikes

Kya Hai: Month ke last 3 days me hospital billing me massive spike dikhana (Ratio > 1.5) taaki monthly targets meet kiye ja sakein.

Kya Nahi Hai: Yeh winter/monsoon diseases seasonality ke peaks nahi hain.

Data Metrics: 5,975 spikes flagged | █229.06 Cr leakage

3. Fraud Ka Financial Impact Aur Audit Ki Recommendations

Patter n	Fraud Type	Cases Flagged	Leakage (Cr)	Audit Action
1a	Card Sharing Concurrent	11,137 pairs	█118.78 Cr	Mandatory biometric scans at entry.
1b	Demographic Mismatch	4,221 claims	█24.40 Cr	Hard validations on gender relationship.
1c	Lazarus Post-Death Billing	524 claims	█0.64 Cr	Real-time integration with Death registry.
2b	Collusion Rings	1,111 rings	█2,700.74 Cr	Audit doctor-hospital referral pairs.
3a	Superman Surgeon	2,193 days	█265.97 Cr	Verify surgeons via AEBAS attendance.
4a	CFA Threshold Avoidance	8,146 claims	█68.13 Cr	Audit claims priced █95k-99.9k.
4b	Duplicate Billing	336,512 pairs	█351.02 Cr	Deduplication block for 72-hour window.
5a	Weekend Emergency	830,866 claims	█4,751.76 Cr	Audit weekend emergency justifications.
5b	Month-End billing spike	5,975 spikes	█229.06 Cr	Audit final week claims for spike hospitals.

"Differentiating between Prevented Leakage (█3,612.41 Cr saved) and Realized Leakage (█3,225.01 Cr slipped) is key. This Hinglish forensic guide provides senior executives with clear audit evidence to enforce stricter rules."